调和函数的对数梯度估计*

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• 欧氏空间$\mathbb{R}^n$上的调和函数的对数梯度估计.

定理 1 设u为$B_1 \subset \mathbb{R}^n$上的正调和函数，则

$$\sup_{B_{1/2}} |D \log u| \leq C,$$

其中C为关于n的正常数.

证明 令$v = \log u$，则需要对v做梯度估计，即对$|Dv|^2$进行估计， 令$w = |Dv|^2$.
计算Δw.

$$\begin{aligned}
\Delta w = \Delta(|Dv|^2) = \sum_{i=1}^{n})(|Dv|^2)_{ii} &= \sum_{i=1}^{n} \left(\sum_{j=1}^{n} (D_j v)^2 \right)_{ii} \\
&= \sum_{i=1}^{n} \left(2 \sum_{j=1}^{n} D_j v \cdot D_{ji} v \right)_i = 2 \sum_{i,j=1}^{n} (D_{ji} v \cdot D_{ji} v + D_j v \cdot D_{jii} v) \\
&= 2 \sum_{i,j=1}^{n} (D_{ij} v)^2 + 2 \sum_{i=1}^{n} D_i v \cdot D_i(\Delta v) = 2 \sum_{i,j=1}^{n} (D_{ij} v)^2 - 2 \sum_{i=1}^{n} D_i v D_i w.
\end{aligned}$$

其中最后一个等号是由于，

$$\begin{aligned}
\Delta v = \Delta(\log u) = \sum_{i=1}^{n} (\log u)_{ii} &= \sum_{i=1}^{n} \left(\frac{u_i}{u} \right)_i = \sum_{i=1}^{n} \frac{u_{ii} \cdot u - u_i \cdot u_i}{u^2} \\
&= \frac{\Delta u}{u} - \frac{|Du|^2}{u^2} = -\frac{|Du|^2}{u^2} = -|Dv|^2 = -w.
\end{aligned}$$

故有，

$$\Delta w + 2 \sum_{i=1}^{n} D_i v D_i w = 2 \sum_{i,j=1}^{n} (D_{ij} v)^2.$$

现在考虑截断函数$\varphi \in C_0^\infty(B_1)$.
计算

$$\begin{aligned}
\Delta(\varphi w) + 2 \sum_{i=1}^{n} D_i v D_i(\varphi w) =& \varphi \Delta w + w \Delta \varphi + 2(D\varphi \cdot Dw) + 2\varphi \sum_{i=1}^{n} D_i v D_i w + 2w \sum_{i=1}^{n} D_i v D_i \varphi \\
=& w \Delta \varphi + \varphi \left(2 \sum_{i,j=1}^{n} (D_{ij} v)^2 - 2 \sum_{i=1}^{n} D_i v D_i w \right) + 2(D\varphi \cdot Dw)
\end{aligned}$$

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$$
\begin{aligned}
&+2\varphi\sum_{i=1}^{n}D_ivD_iw+2w\sum_{i=1}^{n}D_ivD_i\varphi\\
=&w\Delta\varphi+2\varphi\sum_{i,j=1}^{n}(D_{ij}v)^2+2w\sum_{i=1}^{n}D_ivD_i\varphi+4\sum_{i,j=1}^{n}D_i\varphi D_jvD_{ij}v.
\end{aligned}
$$

根据**Cauchy不等式**，有

$$
4|D_i\varphi D_jvD_{ij}v|\le\varphi(D_{ij}v)^2+\frac{4}{\varphi}(D_i\varphi)^2(D_jv)^2.
$$

从而

$$
\begin{aligned}
\Delta(\varphi w)+2\sum_{i=1}^{n}D_ivD_i(\varphi w)\ge&w\Delta\varphi+2\varphi\sum_{i,j=1}^{n}(D_{ij}v)^2+2w\sum_{i=1}^{n}D_ivD_i\varphi-\varphi\sum_{i,j=1}^{n}(D_{ij}v)^2-\frac{4}{\varphi}\sum_{i,j=1}^{n}(D_i\varphi)^2(D_jv)^2\\
=&w\Delta\varphi+\varphi\sum_{i,j=1}^{n}(D_{ij}v)^2+2w\sum_{i=1}^{n}D_ivD_i\varphi-\frac{4}{\varphi}\sum_{i,j=1}^{n}(D_i\varphi)^2(D_jv)^2\\
\ge&-w|\Delta\varphi|+\varphi\sum_{i,j=1}^{n}(D_{ij}v)^2-2|Dv|^2|Dv||D\varphi|-\frac{4}{\varphi}|D\varphi|^2|Dv|^2\\
=&\varphi\sum_{i,j=1}^{n}(D_{ij}v)^2-2|Dv|^3|D\varphi|-\left(|\Delta\varphi|+\frac{4}{\varphi}|D\varphi|^2\right)|Dv|^2.
\end{aligned}
$$

此时选取的φ需要保证$\dfrac{|D\varphi|^2}{\varphi}$有界.

现在对$\displaystyle\sum_{i,j=1}^{n}(D_{ij}v)^2$估计.

$$
\sum_{i,j=1}^{n}(D_{ij}v)^2\ge\sum_{i=1}^{n}(D_{ii}v)^2\ge\frac{1}{n}\left(\sum_{i=1}^{n}D_{ii}v\right)^2=\frac{(\Delta v)^2}{n}=\frac{w^2}{n}.
$$

其中第二个不等号是由**Hölder不等式**保证的，事实上，

$$
\begin{aligned}
&\left(\sum_{i=1}^{n}(D_{ii}v)^2\right)^{1/2}\left(\sum_{i=1}^{n}1^2\right)^{1/2}\ge\sum_{i=1}^{n}D_{ii}v\\
\Longrightarrow&\sqrt{n}\left(\sum_{i=1}^{n}(D_{ii}v)^2\right)^{1/2}\ge\sum_{i=1}^{n}D_{ii}v\\
\Longrightarrow&\sum_{i=1}^{n}(D_{ii}v)^2\ge\frac{1}{n}\left(\sum_{i=1}^{n}D_{ii}v\right)^2.
\end{aligned}
$$

令$\varphi=\eta^4$，其中$\eta\in C_0^\infty(B_1)$，且在$B_{1/2}$上$\eta=1$，则上述估计有

$$
\begin{aligned}
\Delta(\eta^4w)+2\sum_{i=1}^{n}D_ivD_i(\varphi w)\ge&\frac{1}{n}\eta^4w^2-2|Dv|^3|D\eta^4|-\left(|\Delta\eta^4|+\frac{4}{\eta^4}|D\eta^4|^2\right)|Dv|^2\\
=&\frac{\eta^4w^2}{n}-2|Dv|^3|4\eta^3D\eta|-\left(|12\eta^2|D\eta|^2+4\eta^3\Delta\eta|+\frac{4}{\eta^4}|4\eta^3D\eta|^2\right)|Dv|^2\\
\ge&\frac{\eta^4w^2}{n}-8|Dv|^3|\eta^3||D\eta|-(12\eta^2|D\eta|^2+4|\eta^3||\Delta\eta|+64\eta^2|D\eta|^2)|Dv|^2\\
=&\frac{\eta^4w^2}{n}-8|Dv|^3|\eta^3||D\eta|-4\eta^2(19|D\eta|^2+4|\eta||\Delta\eta|)|Dv|^2\\
\ge&\frac{1}{n}\eta^4|Dv|^4-C_1|\eta^3||Dv|^3-C_2\eta^2|Dv|^2.
\end{aligned}
$$

其中最后一个不等号是由于，$\eta \in C_0^\infty(B_1)$，即η为B_1上具有紧支集的光滑函数，从而η的任意阶导数在B_1 上有界.
根据**Cauchy不等式**，有下面估计.

$$\eta^2|Dv|^2 \le \frac{1}{2}\left(\varepsilon_1\eta^4|Dv|^4 + \frac{1}{\varepsilon_1}\right) = \frac{\varepsilon_1}{2}\eta^4|Dv|^4 + \frac{1}{2\varepsilon_1},$$

$$\begin{aligned}
|\eta^3||Dv|^3 &\le \frac{1}{2}\left(\varepsilon_2\eta^4|Dv|^4 + \frac{\eta^2|Dv|^2}{\varepsilon_2}\right) \\
&\le \frac{1}{2}\left(\varepsilon_2\eta^4|Dv|^4 + \frac{1}{\varepsilon_2}\left(\frac{\varepsilon_1}{2}\eta^4|Dv|^4 + \frac{1}{2\varepsilon_1}\right)\right) \\
&= \frac{\varepsilon_2}{2}\eta^4|Dv|^4 + \frac{\varepsilon_1}{4\varepsilon_2}\eta^4|Dv|^4 + \frac{1}{4\varepsilon_1\varepsilon_2}.
\end{aligned}$$

其中$\varepsilon_1, \varepsilon_2$为待确定的常数. 从而，

$$\begin{aligned}
\Delta(\eta^4 w) + 2\sum_{i=1}^n D_i v D_i(\varphi w) \ge& \frac{1}{n}\eta^4|Dv|^4 - C_1\left(\frac{\varepsilon_2}{2}\eta^4|Dv|^2 + \frac{\varepsilon_1}{4\varepsilon_2}\eta^4|Dv|^4 + \frac{1}{4\varepsilon_1\varepsilon_2}\right) - C_2\left(\frac{\varepsilon_1}{2}\eta^4|Dv|^4 + \frac{1}{2\varepsilon_1}\right) \\
=& \frac{1}{n}\eta^4|Dv|^4 - \left(C_1\frac{\varepsilon_2}{2} + C_1\frac{\varepsilon_1}{4\varepsilon_2} + C_2\frac{\varepsilon_1}{2}\right)\eta^4|Dv|^4 - \left(\frac{C_1}{4\varepsilon_1\varepsilon_2} + \frac{C_2}{2\varepsilon_1}\right).
\end{aligned}$$

令$C_1\frac{\varepsilon_2}{2} + C_1\frac{\varepsilon_1}{4\varepsilon_2} + C_2\frac{\varepsilon_1}{2} = \frac{1}{2n}$，即选取的$\varepsilon_1, \varepsilon_2$满足此条件.则

$$\Delta(\eta^4 w) + 2\sum_{i=1}^n D_i v D_i(\varphi w) \ge \frac{1}{2n}\eta^4|Dv|^4 - C.$$

显然C_1, C_2, C均为关于n, η的正常数.
由于$\eta^4 w$在B_1上非负，且在∂B_1附近，$\eta^4 w = 0$，这是由于，$\eta \in C_0^\infty(B_1)$.从而不妨设$x_0 \in B_1$为$\eta^4 w$的极大值点，则

$$D(\eta^4 w)(x_0) = 0, \quad \Delta(\eta^4 w)(x_0) \le 0.$$

故在x_0处， $\eta^4 w^2(x_0) \le C(n, \eta)$.
若$w(x_0) \ge 1$，则

$$\eta^4 w(x_0) \le \eta^4 w^2(x_0) \le C(n, \eta);$$

若$w(x_0) \le 1$，则

$$\eta^4 w(x_0) \le \eta^4(x_0) \le C(\eta),$$

从而 $\eta^4 w(x_0) \le C(n, \eta)$. 由于$x_0$为$\eta^4 w$的极大值点，则

$$\sup_{B_1} \eta^4 w \le C(n, \eta).$$

由η的定义，从而有

$$\sup_{B_{1/2}} w \le C(n, \eta),$$

即

$$\sup_{B_{1/2}} |D \log u| \le C.$$

□

推论 1 (Harnack's Inequality) 设u为$B_1 \subset \mathbb{R}^n$上的正调和函数，则

$$u(x_1) \le Cu(x_2) \quad \forall x_i, x_2 \in B_{1/2},$$

其中C为关于n的正常数.

证明 由于

$$\begin{aligned}\log\frac{u(x_1)}{u(x_2)} &= \int_0^1 \frac{d}{dt}\log u(tx_1+(1-t)x_2)dt \\ &= (x_1-x_2)\int_0^1 D\log u(tx_1+(1-t)x_2)dt,\end{aligned}$$

根据调和函数的对数梯度估计，则

$$\log\frac{u(x_1)}{u(x_2)} \le |x_1-x_2|\int_0^1 |D\log u(tx_1+(1-t)x_2)|dt \le C_1,$$

即

$$\frac{u(x_1)}{u(x_2)} \le e^{C_1} \Longrightarrow u(x_1) \le Cu(x_2).$$

□

• Riemann流形上的调和函数的对数梯度估计.
首先回顾协变导数与协变微分的概念.

定义 1 (协变导数) 设θ 为(r,s)型的张量场，任给1−形式$\{\eta_i\}_{i=1}^r$及向量场$\{Y_j\}_{j=1}^s$，令

$$\begin{aligned}\nabla:\quad \Gamma(TM)\times\Gamma(\otimes^{r,s}TM) &\longrightarrow \Gamma(\otimes^{r,s}(TM)) \\ (X,\theta) &\longmapsto D_X\theta\end{aligned}$$

其中

$$\begin{aligned}\nabla_X\theta(\eta_1,\cdots,\eta_r;Y_1,\cdots,Y_s) =& X(\theta(\eta_1,\cdots,\eta_r;Y_1,\cdots,Y_s)) \\ &-\sum_{i=1}^r \theta(\eta_1,\cdots,\eta_{i-1},\nabla_X\eta_i,\eta_{i+1},\cdots,\eta_r;Y_1,\cdots,Y_s) \\ &-\sum_{j=1}^s \theta(\eta_1,\cdots,\eta_r;Y_1,\cdots,Y_{j-1},\nabla_X Y_j,Y_{j+1},\cdots,Y_s)\end{aligned}$$

这样定义的$\nabla_X\theta$为(r,s)型张量场，称为θ 关于X的**协变导数**.

定义 2 (协变微分) 设θ为(r,s)型张量场，定义$(r,s+1)$型的张量场$\nabla\theta$如下：任给1−形式$\{\eta_i\}_{i=1}^r$及向量场$\{Y_j\}_{j=1}^s, X$，令

$$\nabla\theta(\eta_1,\cdots,\eta_r;Y_1,\cdots,Y_s,X) = \nabla_X\theta(\eta_1,\cdots,\eta_r;Y_1,\cdots,Y_s)$$

这样定义的$\nabla\theta$为$(r,s+1)$型的场张量（张量场），称为θ 的**协变微分**.

现在考虑$(0,0)$型张量场，即函数f.
对于任意的向量场$X\in\Gamma(TM)$，则

$$\nabla f(X) = \nabla_X f = X(f).$$

对于任意的向量场$X,Y\in\Gamma(TM)$，则

$$\begin{aligned}\nabla^2 f(X,Y) &= \nabla(\nabla f)(X,Y) = \nabla_Y(\nabla f)(X) \\ &= Y(\nabla f(X)) - (\nabla f)(\nabla_Y X) \\ &= Y(Xf) - \nabla_Y X(f).\end{aligned}$$

考虑n维欧氏空间$\mathbb{R}^n$.设$X = X^i\dfrac{\partial}{\partial x^i}$，$Y = Y^j\dfrac{\partial}{\partial x^j}$，则

$$\nabla^2 f(X,Y) = (Y^1,\cdots,Y^n)\left(\frac{\partial^2 f}{\partial x^i\partial x^j}\right)\begin{pmatrix} X^1 \\ \vdots \\ X^n\end{pmatrix}.$$

特别地，

$$\nabla^2 f\left(\frac{\partial}{\partial x^i},\frac{\partial}{\partial x^j}\right)=\frac{\partial}{\partial x^i}\left(\frac{\partial}{\partial x^j}f\right)=\frac{\partial^2 f}{\partial x^i\partial x^j}.$$

由于Levi-Civita联络满足对称条件，即对任意的X,Y，有

$$\nabla_X Y-\nabla_Y X-[X,Y]=0.$$

从而

$$X(Yf)-\nabla_X Yf=Y(Xf)-\nabla_Y Xf,$$

即

$$\nabla^2 f(X,Y)=\nabla^2 f(Y,X).$$

对任意的向量场X,Y,Z，定义

$$R(X,Y)Z=\nabla_Y\nabla_X Z-\nabla_X\nabla_Y Z+\nabla_{[X,Y]}Z.$$

特别地，

$$R(X,Y)f=Y(Xf)-X(Yf)+[X,Y]f=0.$$

从而

$$\nabla^2 f(X,Y)-\nabla^2 f(Y,X)=R(X,Y)f.$$

即**Ricci恒等式**.
对于一般的向量场Z，也有相应的**Ricci恒等式.**
由于

$$\nabla^2 Z(X,Y)=\nabla(\nabla Z)(X,Y)=\nabla_Y(\nabla Z)X=\nabla_Y(\nabla_X Z)-\nabla Z(\nabla_Y X)=\nabla_Y\nabla_X Z-\nabla_{\nabla_Y X}Z,$$

则

$$\begin{aligned}\nabla^2 Z(X,Y)-\nabla^2 Z(Y,X)&=\nabla_Y\nabla_X Z-\nabla_X\nabla_Y Z-\nabla_{\nabla_Y X}Z+\nabla_{\nabla_X Y}Z\\&=\nabla_Y\nabla_X Z-\nabla_X\nabla_Y Z+\nabla_{[X,Y]}Z\\&=R(X,Y)Z.\end{aligned}$$

考虑局部计算，有

$$\begin{aligned}&\nabla^2 f\left(\frac{\partial}{\partial x^i},\frac{\partial}{\partial x^j}\right)=\nabla^2 f\left(\frac{\partial}{\partial x^j},\frac{\partial}{\partial x^i}\right)\\\Longrightarrow&\frac{\partial^2 f}{\partial x^j\partial x^i}-\nabla_{\frac{\partial}{\partial x^j}}\frac{\partial}{\partial x^i}f=\frac{\partial^2 f}{\partial x^i\partial x^j}-\nabla_{\frac{\partial}{\partial x^i}}\frac{\partial}{\partial x^j}f,\end{aligned}$$

由于为Levi-Civita，则

$$\nabla_{\frac{\partial}{\partial x^j}}\frac{\partial}{\partial x^i}-\nabla_{\frac{\partial}{\partial x^i}}\frac{\partial}{\partial x^j}=0,$$

从而

$$f_{ij}=f_{ji}.$$

引理 1 设M为完备的*Riemann*流形，$f\in C^3(M)$，任取$p\in M$，$\{x_i\}$为p点的法坐标系，则

$$\Delta|\nabla f|^2(p)=2\sum_{i,j=1}^n|f_{ij}|^2+2\sum_{i,j=1}^n R_{ij}f_if_j+2\sum_{i=1}^n f_i(\Delta f)_i.$$

其中f_i为f相应于$\dfrac{\partial}{\partial x_i}$的协变微分，$R_{ij}$为$M$的*Ricci*张量.

证明 直接计算.

$$\Delta(|\nabla f|^2)=\sum_{j=1}^{n}(|\nabla f|^2)_{jj}=\sum_{j=1}^{n}\left(\sum_{i=1}^{n}f_i^2\right)_{jj}=\sum_{j=1}^{n}\left(2\sum_{i=1}^{n}f_i\cdot f_{ij}\right)_j$$
$$=2\sum_{i,j=1}^{n}(f_{ij})^2+2\sum_{i,j=1}^{n}f_i\cdot f_{ijj}.$$

根据上述分析，有$f_{ij}=f_{ji}$，则$f_{ijj}=f_{jij}$，由 **Ricci恒等式**，知

$$f_{jij}=f_{jji}+R_{ij}f_j,$$

从而

$$\Delta(|\nabla f|^2)=2\sum_{i,j=1}^{n}(f_{ij})^2+2\sum_{i,j=1}^{n}f_i(f_{jji}+R_{ij}f_j)$$
$$=2\sum_{i,j=1}^{n}|f_{ij}|^2+2\sum_{i=1}^{n}f_i(\Delta f)_i+2\sum_{i,j=1}^{n}R_{ij}f_if_j.$$

□

定理 2 (Yau.S.T) 设M为完备的$Riemann$流形，$\dim M=n\geq 2$，$Ric(M)\geq -(n-1)k(k\geq 0)$，$u$为$M$上的正调和函数，则在$M$的任何测地球$B_a(x)$中，有

$$\sup_{B_{a/2}}\frac{|\nabla u|}{u}\leq C\left(\frac{1+ak^{1/2}}{a}\right),$$

其中C为关于n的常数.

证明 令$\phi=\dfrac{|\nabla u|}{u}$，对$\Delta\phi$进行估计.
计算$\Delta\phi$.
由于

$$\phi_i=\left(\frac{|\nabla u|}{u}\right)_i=\frac{|\nabla u|_i\cdot u-|\nabla u|\cdot u_i}{u^2},$$

则

$$\nabla\phi=\frac{\nabla(|\nabla u|)}{u}-\frac{|\nabla u|\cdot\nabla u}{u^2}.$$

由于$|\nabla u|=u\phi$，则

$$\Delta(|\nabla u|)=\Delta(u\cdot\phi)=u\Delta\phi+\phi\Delta u+2\nabla\phi\cdot\nabla u=u\Delta\phi+2\nabla\phi\cdot\nabla u,$$

从而

$$\Delta\phi=\frac{\Delta(|\nabla u|)}{u}-2\frac{\nabla\phi\cdot\nabla u}{u}=\frac{|\nabla u|\Delta(|\nabla u|)}{|\nabla u|u}-2\frac{\nabla\phi\cdot\nabla u}{u}.$$

现在需要对$|\nabla u|\Delta(|\nabla u|)$进行估计.

计算$\Delta(|\nabla u|^2)$.
依**引理1**，在$B_a(x)$上一点p处，有

$$\Delta(|\nabla u|^2)=2\sum_{i,j=1}^{n}(u_{ij})^2+2\sum_{i=1}^{n}u_i(\Delta u)_i+2Ric(\nabla u,\nabla u)$$
$$\geq 2\sum_{i,j=1}^{n}(u_{ij})^2-2(n-1)k|\nabla u|^2.$$

由于

$$\Delta(|\nabla u|^2) = 2|\nabla u|\Delta(|\nabla u|) + 2|\nabla|\nabla u||^2,$$

则

$$|\nabla u|\Delta(|\nabla u|) \geq \sum_{i,j=1}^{n} (u_{ij})^2 - (n-1)k|\nabla u|^2 - |\nabla|\nabla u||^2.$$

计算$|\nabla|\nabla u||^2$.
选取p点法坐标系，使得$u_i(p) = 0(i \geq 2)$，$u_1(p) = |\nabla u|_{(p)}$.
由于

$$(|\nabla u|)_j = \frac{1}{2} \cdot \frac{1}{|\nabla u|} \cdot 2\sum_{i=1}^{n} u_i u_{ij} = u_{1j},$$

则

$$|\nabla|\nabla u||^2 = \sum_{j=1}^{n} u_{1j}^2.$$

则

$$\begin{aligned}
|\nabla u|\Delta(|\nabla u|) + (n-1)k|\nabla u|^2 &\geq \sum_{i,j=1}^{n} u_{ij}^2 - \sum_{j=1}^{n} u_{1j}^2 \\
&\geq \sum_{i=2}^{n} (u_{i1})^2 + \sum_{i=2}^{n} u_{ii}^2 \\
&\geq \sum_{i=2}^{n} (u_{i1})^2 + \frac{1}{n-1}\left(\sum_{i=2}^{n} u_{ii}\right)^2.
\end{aligned}$$

由于$\Delta u = 0$，则$u_{11}^2 = \left(\sum_{i=2}^{n} u_{ii}\right)^2$， 从而

$$|\nabla u|\Delta(|\nabla u|) + (n-1)k|\nabla u|^2 \geq \frac{1}{n-1}\sum_{i=1}^{n} u_{i1}^2 = \frac{1}{n-1}|\nabla|\nabla u||^2.$$

回到对$\Delta\phi$的估计.

$$\begin{aligned}
\Delta\phi &= \frac{|\nabla u|\Delta(|\nabla u|)}{|\nabla u|u} - 2\frac{\nabla\phi \cdot \nabla u}{u} \\
&\geq \frac{1}{|\nabla u|u}\left(\frac{1}{n-1}|\nabla|\nabla u||^2 - (n-1)k|\nabla u|^2\right) - 2\frac{\nabla\phi \cdot \nabla u}{u} \\
&= \frac{1}{(n-1)|\nabla u|u}|\nabla|\nabla u||^2 - (n-1)k\phi - 2\frac{\nabla\phi \cdot \nabla u}{u}.
\end{aligned}$$

现在对$\dfrac{\nabla\phi \cdot \nabla u}{u}$进行估计.
考虑$\varepsilon = \dfrac{2}{n-1}$，则

$$\begin{aligned}
-2\frac{\nabla\phi \cdot \nabla u}{u} &= -(2-\varepsilon)\frac{\nabla\phi \cdot \nabla u}{u} - \varepsilon\frac{\nabla\phi \cdot \nabla u}{u} \\
&= -(2-\varepsilon)\frac{\nabla\phi \cdot \nabla u}{u} - \varepsilon\left(\frac{\nabla u \cdot \nabla(|\nabla u|)}{u^2} - \frac{|\nabla u|^3}{u^3}\right) \\
&\geq -(2-\varepsilon)\frac{\nabla\phi \cdot \nabla u}{u} - \varepsilon\frac{|\nabla u||\nabla(|\nabla u|)|}{u^2} + \varepsilon\phi^3
\end{aligned}$$

$$
\begin{aligned}
&\geq -(2-\varepsilon)\frac{\nabla\phi\cdot\nabla u}{u} - \varepsilon\left(\frac{|\nabla(|\nabla u|)|}{(|\nabla u|u)^{1/2}}\cdot\frac{|\nabla u|^{3/2}}{u^{3/2}}\right) + \varepsilon\phi^3 \\
&\geq -(2-\varepsilon)\frac{\nabla\phi\cdot\nabla u}{u} - \frac{\varepsilon}{2}\left(\frac{|\nabla(|\nabla u|)|^2}{|\nabla u|u} + \phi^3\right) + \varepsilon\phi^3 \\
&= -(2-\varepsilon)\frac{\nabla\phi\cdot\nabla u}{u} - \frac{1}{n-1}\left(\frac{|\nabla(|\nabla u|)|^2}{|\nabla u|u} + \phi^3\right) + \varepsilon\phi^3,
\end{aligned}
$$

从而

$$
\Delta\phi \geq -(n-1)k\phi - \left(2-\frac{2}{n-1}\right)\frac{\nabla\phi\cdot\nabla u}{u} + \frac{1}{n-1}\phi^3.
$$

考虑$B_a(x)$的函数

$$
F(y) = (a^2-\rho^2)\phi(y) = (a^2-\rho^2)\frac{|\nabla u|}{u},
$$

其中$\rho = d(y,x)$.显然$F\mid_{\partial B_a} = 0$，若$\nabla u! = 0$，则F在$B_a(x)$内部达到极大值，在极大值点处$\nabla u \neq 0$，若$\nabla u = 0$，则结论显然.
设$x_1 \in B_a(x)$为$F(y)$的极大值点.若x_1不为x的割点，则有极大值原理，有

$$
\begin{cases}
\nabla F(x_1) = 0 \\
\Delta F(x_1) \leq 0.
\end{cases}
$$

直接计算：

$$
\nabla F(y) = -\nabla\rho^2\phi(y) + (a^2-\rho^2)\nabla\phi(y),
$$

$$
\Delta F(y) = \Delta((a^2-\rho^2)\phi(y)) = (a^2-\rho^2)\Delta\phi(y) - \Delta\rho^2\cdot\phi(y) - 2\nabla\rho^2\cdot\nabla\phi(y).
$$

从而在x_1处，有

$$
\begin{cases}
\dfrac{\nabla\rho^2}{a^2-\rho^2} = \dfrac{\nabla\phi}{\phi}, \\
-\dfrac{\Delta\rho^2}{a^2-\rho^2} + \dfrac{\Delta\phi}{\phi} - \dfrac{2\nabla\rho^2\cdot\nabla\phi}{(a^2-\rho^2)\phi} \leq 0.
\end{cases}
$$

从而

$$
\frac{\Delta\phi}{\phi} - \frac{\Delta\rho^2}{a^2-\rho^2} - \frac{2|\nabla\rho^2|^2}{(a^2-\rho^2)^2} \leq 0.
$$

由于$|\nabla\rho^2| = 2\rho|\nabla\rho| = 2\rho$，且由比较定理知

$$
\begin{aligned}
\Delta\rho^2 &= 2\rho\Delta\rho + 2|\nabla\rho|^2 \\
&= 2 + 2\rho\Delta\rho \\
&\leq 2 + 2(n-1)(1+\sqrt{k}\rho) \\
&\leq C(1+\sqrt{k}\rho),
\end{aligned}
$$

其中C关于n，从而

$$
0 \geq \frac{\Delta\phi}{\phi} - \frac{C(1+\sqrt{k}\rho)}{a^2-\rho^2} - \frac{8\rho^2}{(a^2-\rho^2)^2}
$$

结合$\Delta\phi$的估计，有

$$
0 \geq -(n-1)k - \left(2-\frac{2}{n-1}\right)\frac{\nabla\phi\cdot\nabla u}{\phi u} + \frac{1}{n-1}\phi^2 - \frac{C(1+\sqrt{k}\rho)}{a^2-\rho^2} - \frac{8\rho^2}{(a^2-\rho^2)^2}.
$$

由于$-\dfrac{\nabla\phi\cdot\nabla u}{\phi u} = -\dfrac{2\rho\nabla\rho\cdot\nabla u}{(a^2-\rho^2)u} \geq -\dfrac{2\rho}{a^2-\rho^2}\phi$，则

$$
0 \geq -(n-1)k - \left(2-\frac{2}{n-1}\right)\frac{2\rho}{a^2-\rho^2}\phi + \frac{1}{n-1}\phi^2 - \frac{C(1+\sqrt{k}\rho)}{a^2-\rho^2} - \frac{8\rho^2}{(a^2-\rho^2)^2}.
$$

直接计算有，

$$
\begin{aligned}
0 &\geq \frac{1}{n-1}F^2 - \frac{4(n-2)}{n-1}\rho F - C(1+\sqrt{k}\rho)(a^2-\rho^2) - 8\rho^2 - k(a^2-\rho^2)^2 \\
&\geq \frac{1}{n-1}F^2 - 2C_1 aF - C(1+\sqrt{k}\rho)a^2 - 8a^2 - ka^4 \\
&\geq \frac{1}{n-1}F^2 - 2C_1 aF - C_2(1+\sqrt{k}a)^2 a^2.
\end{aligned}
$$

其中C_1, C_2只与n相关.则

$$F(x_1) = \sup_{B_a} F \leq C_n a(1+\sqrt{k}a).$$

从而

$$\frac{3}{4}a^2 \sup_{B_{a/2}} \frac{|\nabla u|}{u} \leq C_n a(1+\sqrt{k}a),$$

即

$$\sup_{B_{a/2}} \frac{|\nabla u|}{u} \leq C_n \left(\frac{1+\sqrt{k}a}{a}\right).$$

□

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